

LAB WORKSHEET

Your First Cloud Security Honeypot

Objective: Build a vulnerable virtual machine in the cloud, expose it to the internet, and use enterprise SIEM tools (Microsoft Sentinel) to track live cyber attacks from across the globe.

Phase 1: The Funding

We need infrastructure. Let's use Microsoft's money to build it. Microsoft Azure is a leading public cloud platform. By using your student status, you gain access to a sandbox environment without risking personal funds.

- ☐ Navigate to: azure.microsoft.com/en-us/free/students/
- ☐ Authenticate using your `@student.unimelb.edu.au` email address. Azure uses this domain to verify your active student status automatically.
- ☐ Claim your \$100 free compute credits. This process requires zero credit card details, ensuring you cannot accidentally be billed.

Phase 2: The Bait (Virtual Machine)

We are building the trap. A Virtual Machine (VM) is an emulation of a physical computer. We want attackers to find this, but we don't want them to actually get in.

- ☐ In the Azure Portal search bar, type **Virtual Machines** and click **Create**.
- ☐ **Resource Group:** Create new → Name it `MISC_HoneyPot_Lab`. A resource group is a logical container holding related resources for a single project.
- ☐ **Virtual machine name:** `win-honeyPot-01`
- ☐ **Region:** Select any region (e.g., Australia East). This dictates the physical datacenter where your server lives.
- ☐ **Image:** Select **Windows 10 Pro** or **Windows Server 2022**. We use Windows because its login failures are highly standardized in Event Logs.
- ☐ **Size:** `Standard_B1s`. This is a low-cost instance (1 vCPU, 1GB RAM) perfect for our lightweight honeypot.
- ☐ **Administrator Account:**
 - **Username:** _____ (Avoid generic names like "admin")
 - **Password:** Create a 20+ character password (e.g., a long phrase).
CRITICAL WARNING: Bots WILL try to guess this via brute-force. If they succeed, your VM will be hijacked for crypto-mining and your Azure account will be banned. Do not use a simple password!
- ☐ **Inbound Port Rules:** Select "Allow selected ports" → Check **RDP (3389)**. RDP (Remote Desktop Protocol) allows remote graphical access to the system. Leaving it open to the public internet (0.0.0.0/0) is our trap.

- ☐ Click **Review + Create**, wait for validation to pass, then click **Create**.

Lab Record: Once deployed, click "Go to resource" to find the public IP address of your VM.
My Honeypot Public IP is: _____

Phase 3: The Trap (Microsoft Sentinel)

Now we set up the surveillance system. Microsoft Sentinel is a scalable, cloud-native SIEM (Security Information and Event Management) system. It ingests logs from our VM and analyzes them.

- ☐ In the Azure search bar, type **Microsoft Sentinel**.
- ☐ Click **Create Microsoft Sentinel**.
- ☐ Click **Create a new workspace** → Name it MISC-Sentinel-Workspace. A Log Analytics Workspace is the underlying database where your logs will be stored and queried.
- ☐ Once the workspace is created, select it and click **Add** to attach it to Sentinel.
- ☐ Inside Sentinel, navigate to the left menu and select **Data Connectors**.
- ☐ Search for **Windows Security Events via AMA** (Azure Monitor Agent). The AMA is a small service installed on the VM that forwards its logs to our workspace.
- ☐ Click **Open connector page** → **Create Data Collection Rule**.
- ☐ Name it Honeypot-Logs. Under the "Resources" tab, select your win-honeypot-01 VM.
- ☐ Under "Collect", select **"All Security Events"** and deploy.

Phase 4: The Validation

Take a break. Within the next 1-4 hours, automated scanners from across the globe will find your open RDP port and attempt to log in using stolen password dictionaries.

- ☐ Open Microsoft Sentinel → **Logs**.
- ☐ Close the default pop-up to access the empty query window.
- ☐ Run this KQL (Kusto Query Language) script to parse the security events. **EventID 4625** specifically maps to "An account failed to log on" in Windows.

```
SecurityEvent
| where EventID == 4625 // Filters for failed authentication events
| summarize count() by IPAddress, TargetUserName // Groups attempts by attacker IP
| sort by count_ desc // Orders the results from highest attempts to lowest
```

Lab Findings: Run the query and analyze your logs.

1. Which IP address attacked you the most? _____
2. What was the most common username they guessed? _____
3. Total number of failed login attempts so far: _____

Phase 5: The HR Bypass

Turn this lab into proof of your skills for future employers. A well-documented project shows initiative and technical communication skills.

- ☐ Take a clear screenshot of your Microsoft Sentinel logs showcasing the failed login attempts.
- ☐ Create a new public repository on your GitHub account (e.g., `azure-sentinel-honeypot`).
- ☐ Upload the screenshot and write a comprehensive `README.md` detailing:
 - **Architecture:** What you built (Azure VM, Log Analytics Workspace, Sentinel, Data Collection Rules).
 - **Objective:** Why you built it (To observe and analyze live brute-force RDP attacks on an exposed honeypot).
 - **Skills Demonstrated:** What you learned (KQL querying, SIEM deployment, cloud infrastructure provisioning).
- ☐ Add the GitHub repository link to the "Projects" section of your resume.

Phase 6: The Clean Up (DO NOT SKIP!)

Cloud engineers never pay for what they aren't using. The cloud uses an OpEx (Operational Expenditure) model—you are billed by the minute. Leaving this running will drain your \$100 credit.

- ☐ Search for **Resource Groups** in the Azure search bar.
- ☐ Click on your resource group: `MISC_Honeypot_Lab`.
- ☐ Click **Delete resource group** at the top of the page.
- ☐ Type the exact name of the group (`MISC_Honeypot_Lab`) to confirm.
- ☐ Click **Delete**. *Deleting the resource group performs a cascading delete, safely destroying the VM, Sentinel workspace, virtual network, and all hidden costs in one action.*